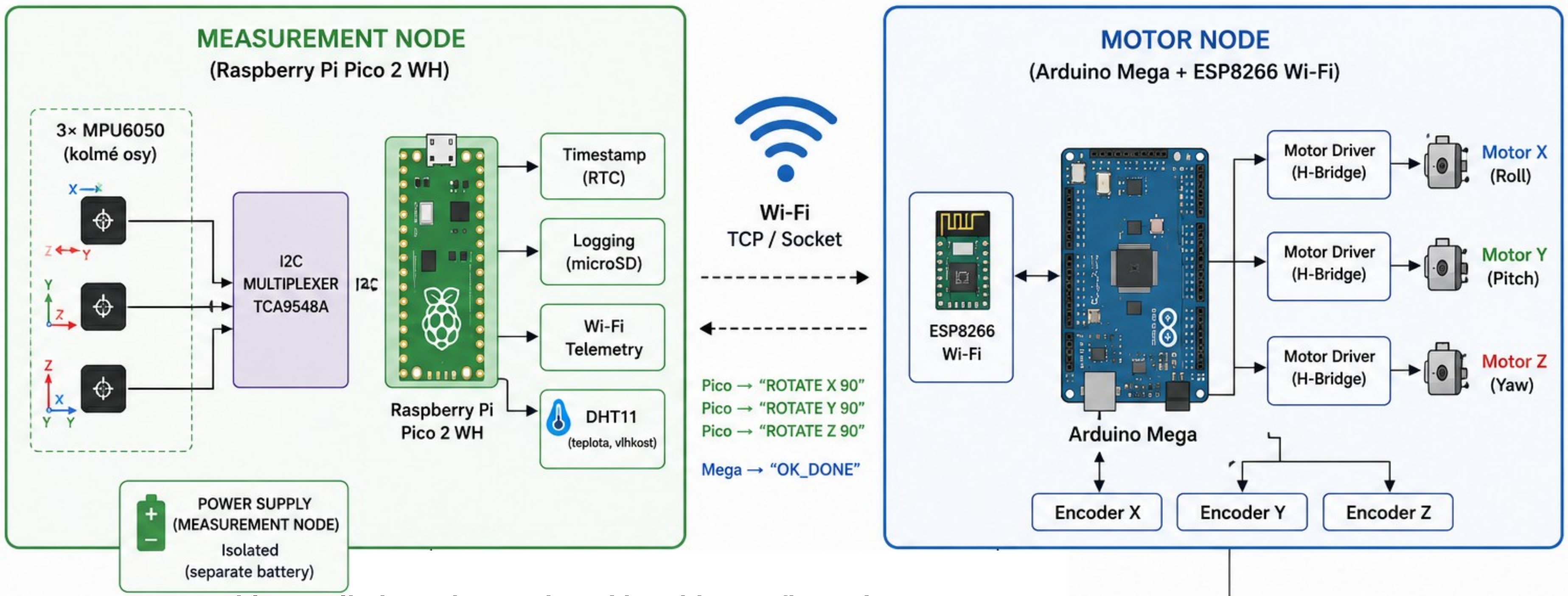

Problems Eliminated or Reduced by This Configuration

Ground loop interference
PWM switching noise coupling into IMU electronics
Voltage spikes from DC motors
Mechanical cable pulling artifacts
Slip ring electrical noise
Crosstalk between motor electronics and measurement electronics
Electromagnetic contamination from H-bridges
Timing instability caused by motor control interrupts
Microphonic vibration artifacts inside MEMS structures
Random motion induced by dangling wires
Electrical transients during motor startup/shutdown
Noise injection through common GND reference
Data corruption caused by noisy motor environments
Mechanical resonance amplification through shared chassis elements
Reduced long-term stability caused by mixed analog/digital noise sources
Environmental contamination of measurements caused by uncontrolled artifacts
Improved statistical purity of long-term MEMS measurements
Improved platform symmetry and inertial balance
Reduced probability of interpreting actuator artifacts as physical phenomena
Reduced conducted noise inside sensitive IMU circuitry
Reduced structural vibration propagation into the gyro platform
Improved suitability for Allan variance measurements
Improved suitability for ultra-low-frequency signal analysis

EMI generated by motor drivers
Shared power supply ripple
Current surge induced sensor instability
Rotating cable torsion effects
USB ground coupling
Vibrations transferred through rigid cable harnesses
ADC noise caused by shared motor power rails
Sensor drift amplification caused by unstable supply voltage
Mechanical asymmetry caused by external cable drag
Thermal coupling between power electronics and sensors
High-frequency switching harmonics from PWM drivers
Synchronization errors caused by mixed control and measurement tasks
Magnetic field disturbances generated by nearby motors
Partial elimination of non-measurement-related systematic actuator activity
Improved separation between measurement domain and actuation domain
Improved repeatability of experimental conditions
Reduced risk of false correlated signals between sensor axes
Reduced influence of power-stage thermal drift on sensor readings
Improved conditions for FFT analysis and long-term drift analysis
Improved separation of local artifacts from potentially global/systemic effects